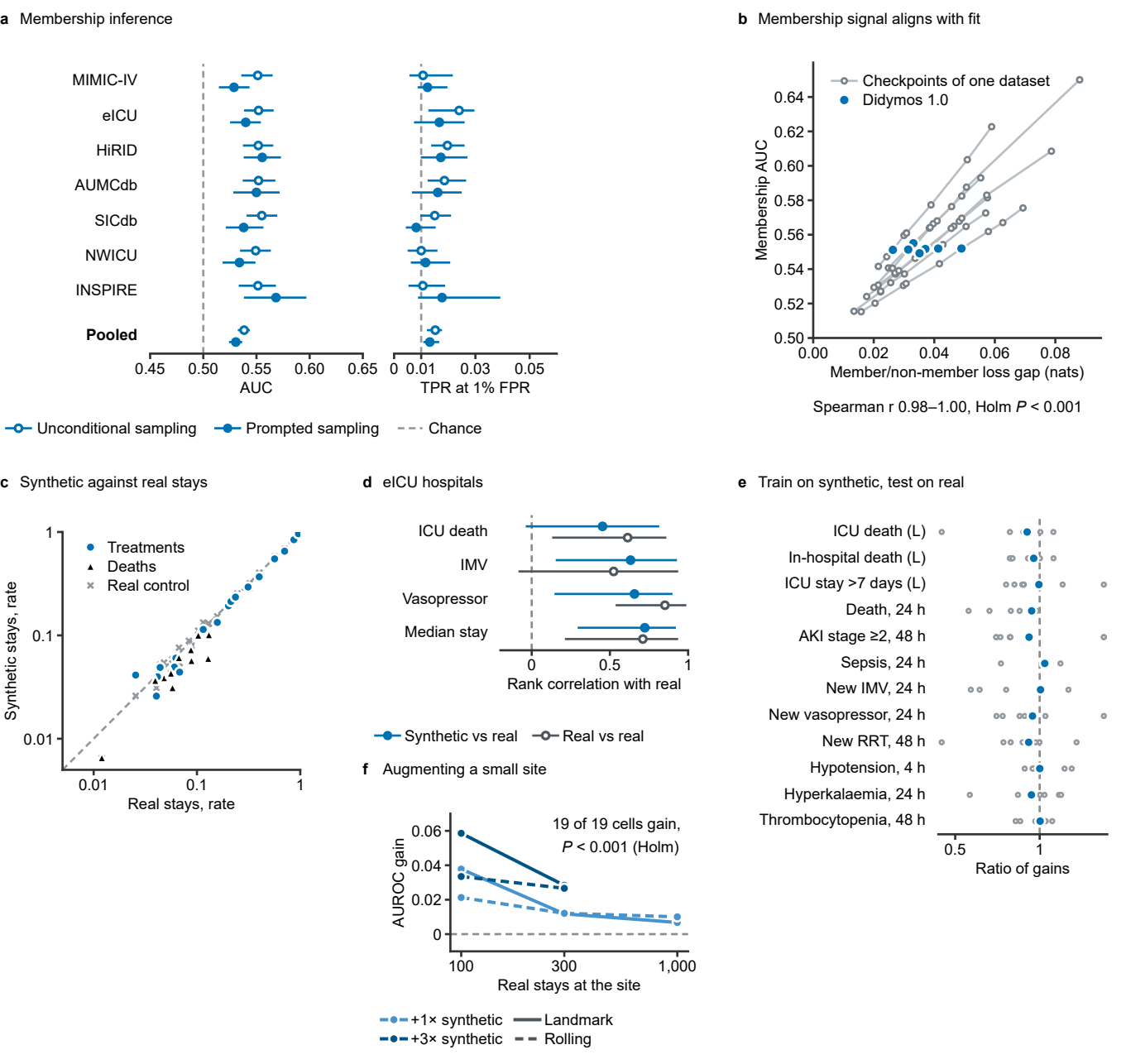

a Membership inference
MIMIC-IV
eICU
HiRID
AUMCdb
SICdb
NWICU
INSPIRE
Pooled
0.45 0.50 0.55 0.60 0.65
AUC
0 0.01 0.03 0.05
TPR at 1% FPR
Unconditional sampling
Prompted sampling
Chance
b Membership signal aligns with fit
Checkpoints of one dataset
Didymos 1.0
Membership AUC
0.50 0.52 0.54 0.56 0.58 0.60 0.62 0.64
0.00 0.02 0.04 0.06 0.08
Member/non-member loss gap (nats)
Spearman r 0.98–1.00, Holm P < 0.001
c Synthetic against real stays
Treatments
Deaths
Real control
Synthetic stays, rate
1 0.1 0.01
0.01 0.1 1
Real stays, rate
d eICU hospitals
ICU death
IMV
Vasopressor
Median stay
0 0.5 1
Rank correlation with real
Synthetic vs real
Real vs real
e Train on synthetic, test on real
ICU death (L)
In-hospital death (L)
ICU stay >7 days (L)
Death, 24 h
AKI stage ≥2, 48 h
Sepsis, 24 h
New IMV, 24 h
New vasopressor, 24 h
New RRT, 48 h
Hypotension, 4 h
Hyperkalaemia, 24 h
Thrombocytopenia, 48 h
0.5 1
Ratio of gains
f Augmenting a small site
19 of 19 cells gain,
P < 0.001 (Holm)
AUROC gain
0.06 0.04 0.02 0
100 300 1,000
Real stays at the site
+1× synthetic
+3× synthetic
Landmark
Rolling